Dear Hiring Manager,

I am applying for the Machine Learning Engineer – Customer Support AI role at Quillstone Payments. I have more than five years of experience in NLP and document classification, spanning research, prototyping, and production machine-learning systems. I am interested in applying that work to systems that help merchants and support agents. I hold a Master of Computer Science and am completing a PhD in Computer Science focused on named entity recognition in Croatian legal texts.

At Slavonica Document AI, I developed an XLM-R classifier for public administration documents that achieved 94% accuracy across 23 classes and now processes approximately 15,000 documents a day. Using Python, PyTorch, and Hugging Face Transformers, I introduced MLflow for experiment tracking and model registry, deployed FastAPI model-serving services in Docker on Azure ML and Azure App Service, and established production drift monitoring. I also hold the Microsoft Certified: Azure AI Engineer Associate certification (2024). That monitoring detected an OCR change that had reduced accuracy by nine percentage points; the issue was fixed within a week. I also mentor two student interns.

At Baranjska kreditna banka, I engineered TF-IDF text features and classified customer complaints into 11 topics using logistic regression, followed by fastText, supporting weekly reporting for customer service management. I have also fine-tuned BERTić for Croatian named entity recognition and built an annotated dataset of 4,000 documents with two annotators. My doctoral research, currently in progress, focuses on named entity recognition in Croatian legal texts. I also built a personal retrieval-augmented generation (RAG) question-answering prototype over Croatian laws using sentence embeddings, FAISS, and an open-weight LLM accessed through an API; it was evaluated through manual checks and was not used in production.

Quillstone's focus on taking support AI from prototype to production aligns with the work I have done across model development, deployment and monitoring. I would welcome the opportunity to discuss how my NLP and production experience could contribute to your Customer Operations AI team.

Kind regards,
Petra Zlatarić-Novosel